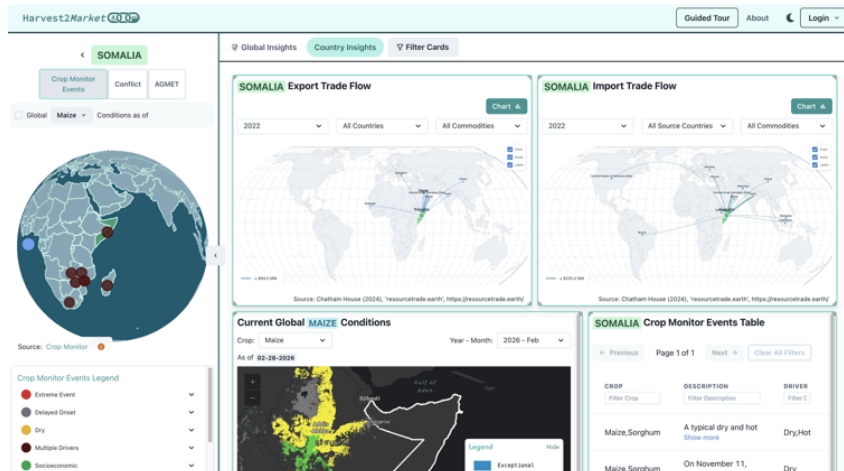


Somalia - Harvest2Market Visuals & Datasets

This document outlines files contained in the **Somalia_Harvest2Market** folder. The files contained in this folder are available in the Harvest2Market Country Insights page for Somalia, accessible via link:

<https://harvest2market.nasaharvest.org/som>



Not all cards have data for Somalia, but for those available, this documentation includes the JavaScript files, required libraries, and links to the resources available through the Harvest portal or directly from sources themselves (e.g., FEWS NET, Chatham House / ResourceTrade.earth). For resources that require an API key, this will be noted where you will need to obtain one yourself.

Note: The code provided was written to work within the Harvest2Market codebase and will require edits to paths so it works with your new codebase.

Parent Folder: Trade Flow

File: TradeMapExports.js

Required Libraries

Data Sources

API Reference - ResourceTrade.earth API

Local Data Files Required

Component Imports (found in sub-folder "imports")

File: TradeMapImports.js

Required Libraries

Data Sources

API Reference - ResourceTrade.earth API

Local Data Files Required

Component Imports (found in sub-folder "imports")

Parent Folder: CropMonitor

File: CountryCMEventTable.js

Required Libraries

Data Sources

API Reference - GEOGLAM Crop Monitor Events API

Local Data Files Required

Component Imports (found in sub-folder "imports")

File: CountryCropConditions.js

Required Libraries

Data Sources

API Reference - GEOGLAM Crop Monitor Map Services

Local Data Files Required

Component Imports (found in sub-folder "imports")

Parent Folder: FEWSNET

File: AcuteFoodInsecurityMap.js

Required Libraries

Data Sources

API Reference

Harvest Portal API

Admin Boundary Resource IDs:

FEWS NET API (via Lambda Proxy)

Local Data Files Required

Country Resource Map

File: IPCPopSizeChart.js

Required Libraries

Data Sources

API Reference

FEWS NET API (via Lambda Proxy)

Local Data Files Required

File: MarketPriceFacts.js

Data Sources

API Reference

Local Data Files Required

Parent Folder: Trade Flow

Trade Flow/

└─ TradeMapExports.js

└─ TradeMapImports.js

└─ imports/

| └─ SpatialSankey.js

| └─ SpatialSankeyImports.js

└─ data/

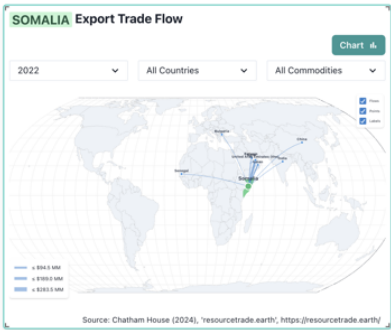
└─ faostat-country-codes.json

└─ iso-3166-country-codes.json

└─ geoBoundariesCGAZ_ADM0.json

└─ cgaz_adm0_simple.json

File: TradeMapExports.js



Map View

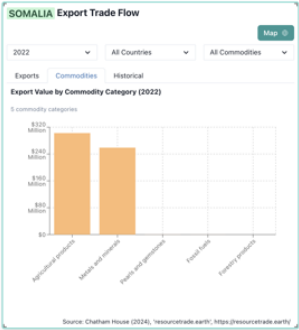


Chart View

Required Libraries

```
1 {  
2   "dependencies": {  
3     ...  
4   }  
5 }
```

```

3    "@chakra-ui/react": "^2.x",
4    "@turf/area": "^6.x",
5    "@turf/center-of-mass": "^6.x",
6    "react": "^18.x",
7    "react-icons": "^4.x",
8    "react-redux": "^8.x",
9    "react-simple-maps": "^3.x",
10   "recharts": "^2.x",
11   "topojson-client": "^3.x"
12 }
13 }

```

Data Sources

Resource	Source	API Key Required
Trade Data	ResourceTrade.earth API	No
Country Codes	Local JSON files	No
Geographic Boundaries	Local TopoJSON files	No

API Reference - ResourceTrade.earth API

Base URL: `https://api.resourcetrade.earth/api/rt/2.6/`

No API key required for the ResourceTrade.earth API.

Endpoints Used

Endpoint	Description	Parameters
<code>/trades</code>	Get trade flow data	year, exporter, importer, category
<code>/commodities</code>	Get commodity breakdown	year, exporter, importer, category
<code>/historical</code>	Get historical trade data	year, exporter, importer, category
<code>/models</code>	Get category/commodity definitions	None

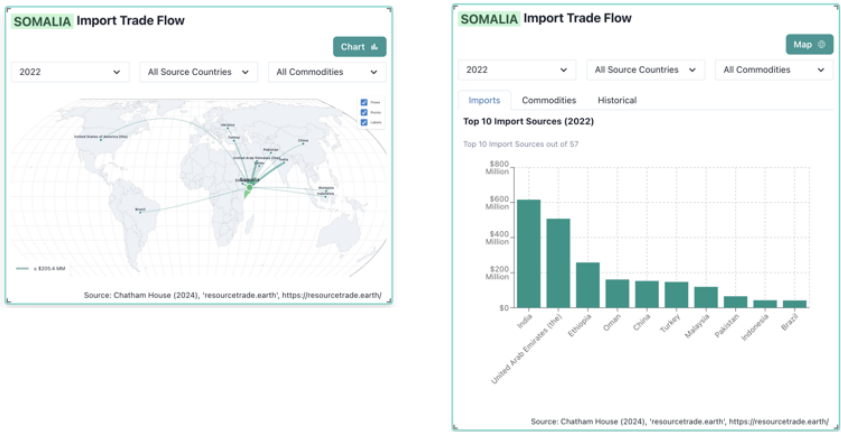
Local Data Files Required

- `faostat-country-codes.json` - Country code mappings
- `iso-3166-country-codes.json` - ISO country codes
- `geoBoundariesCGAZ_ADM0.json` - Country boundaries (TopoJSON)
- `cgaz_adm0_simple.json` - Simplified country boundaries

Component Imports (found in sub-folder "imports")

- `SpatialSankey.js` - Flow visualization component
 - Ignore: `ciLayout.js` - Layout configuration utility not essential
-

File: TradeMapImports.js



Required Libraries

Same as TradeMapExports.js (see above)

Data Sources

Same as TradeMapExports.js (see above)

API Reference - ResourceTrade.earth API

Same as TradeMapExports.js (see above)

Local Data Files Required

Same as TradeMapExports.js (see above)

Component Imports (found in sub-folder "imports")

- SpatialSankeyImports.js - Import flow visualization component
- Ignore file: ciLayout.js - not essential

Parent Folder: CropMonitor

CropMonitor/
├ CountryCMEventTable.js
├ CountryCropConditions.js
├ imports/
| └ Legends.js

| └ footnotes.js
├ data/
| └ faostat-country-codes.json

File: CountryCMEventTable.js

This component displays a filterable, sortable table of crop monitor events reported for Somalia, including information about crop conditions, drivers of change, and impacts.

SOMALIA Crop Monitor Events Table						
Previous Page 1 of 1 Next Clear All Filters						
CROP	DESCRIPTION	DRIVER	IMPACT	COUNTRY	SOURCE	START DATE
Filter Crop	Filter Description	Filter C	All	Filter Country	Filter S	Filter Start Date
Maize,Sorghum	A typical dry and hot January Show more	Dry,Hot	Major Negative	Somalia	Link	02/19/2026
Maize,Sorghum	On November 11, the Show more	Dry	Minor Negative	Somalia	Link	11/11/2025
Maize,Sorghum	https://fews.net/east- Show more		Minor Positive	Somalia	Link	09/08/2025
Maize, Sorghum	Forecast below-average Show more	Dry		Somalia	Link	10/01/2024
Showing 4 of 4 results						

Required Libraries

```
1 {
2   "dependencies": {
3     "@chakra-ui/react": "^2.x",
4     "@chakra-ui/icons": "^2.x",
5     "@tanstack/react-table": "^8.x",
6     "react": "^18.x",
7     "react-redux": "^8.x"
8   }
9 }
```

Data Sources

Resource	Source	API Key Required
Crop Monitor Events	ArcGIS Feature Service	No

API Reference - GEOGLAM Crop Monitor Events API

Base URL: `https://services7.arcgis.com/WSiUmUh1Fx4CtMBB/ArcGIS/rest/services/`

No API key required for this ArcGIS Feature Service.

Endpoint Used

Endpoint	Description	Parameters
<code>service_e5fc4330cb5340a59abc6c7d4ae2e46d/FeatureServer/0/query</code>	Query crop monitor events	<code>f=json</code> , <code>where</code> , <code>outFields</code>

Example Request:

```
1 https://services7.arcgis.com/WSiUmUh1Fx4CtMBB/ArcGIS/rest/services/service_e5fc4330cb5340a59abc6c7d4ae2e46d/FeatureServer/0/query?f=json&where=1=1&outFields=*
```

Local Data Files Required

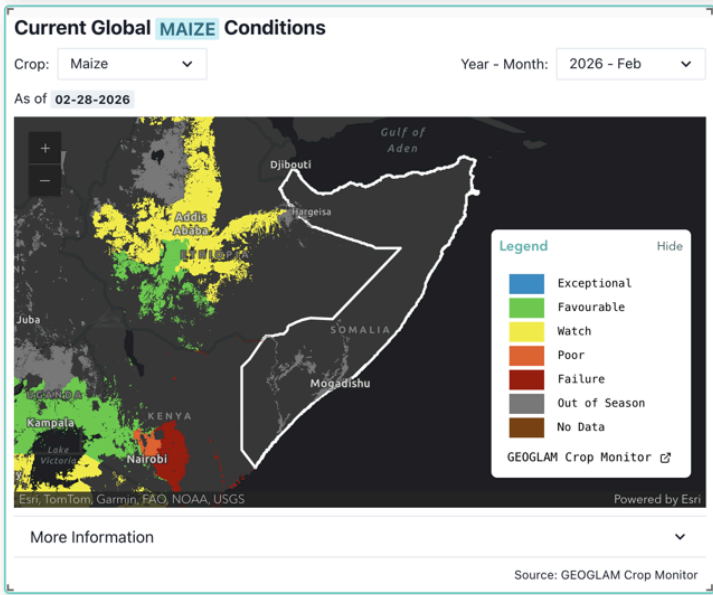
- `faostat-country-codes.json` - Country code mappings (ISO2, ISO3)

Component Imports (found in sub-folder "imports")

- `footnotes.js` - Footnote definitions for Crop Monitor events

File: CountryCropConditions.js

This component displays an interactive map showing current global crop conditions with the ability to filter by crop type (wheat, maize, rice, soybean, sorghum, millet) and time period. The selected country (Somalia) is highlighted on the map.



Required Libraries

```
1 {
2   "dependencies": {
3     "@arcgis/core": "^4.x",
4     "@chakra-ui/react": "^2.x",
5     "react": "^18.x",
6     "react-redux": "^8.x"
7   }
8 }
```

Note: The ArcGIS JavaScript API (@arcgis/core) requires a valid ArcGIS API key or ArcGIS Online account for some basemap styles. The dark-gray-vector basemap used in this component is available without authentication for development purposes.

Data Sources

Resource	Source	API Key Required
Crop Condition Layers	GEOGLAM Crop Monitor MapServer	No
Country Boundaries	ArcGIS GEOGLAM Feature Service	No

API Reference - GEOGLAM Crop Monitor Map Services

No API key required for these services.

Base URLs:

Crop	MapServer URL
------	---------------

Wheat	<code>https://data.cropmonitor.org/arcgis/rest/services/CMET/Wheat/MapServer</code>
Maize	<code>https://data.cropmonitor.org/arcgis/rest/services/CMET/Maize/MapServer</code>
Rice	<code>https://data.cropmonitor.org/arcgis/rest/services/CMET/Rice/MapServer</code>
Soybean	<code>https://data.cropmonitor.org/arcgis/rest/services/CMET/Soybean/MapServer</code>
Sorghum	<code>https://data.cropmonitor.org/arcgis/rest/services/CMET/Sorghum/MapServer</code>
Millet	<code>https://data.cropmonitor.org/arcgis/rest/services/CMET/Millet/MapServer</code>

Country Boundaries:

```
1 | https://services1.arcgis.com/qTQ6qYkHpXlu0G82/ArcGIS/rest/services/GE06/LAM_Crop_Monitor_Map_Layers/FeatureServer/7
```

Example Request (Get available layers):

```
1 | https://data.cropmonitor.org/arcgis/rest/services/CMET/Maize/MapServer?f=pjson
```

Local Data Files Required

File	Description	Location
<code>faostat-country-codes.json</code>	Country code mappings (ISO2, ISO3)	data/

Component Imports (found in sub-folder "imports")

Import	Description
<code>Legends.js</code>	ConditionsLegend component for map legend
<code>footnotes.js</code>	Condition definitions (Exceptional, Favourable, Watch, Poor, Failure)

Parent Folder: FEWSNET

FEWSNET/

- |— AcuteFoodInsecurityMap.js
- |— IPCPopSizeChart.js
- |— MarketPriceFacts.js
- └─ data/
 - |— faostat-country-codes.json

AWS Backend Configuration

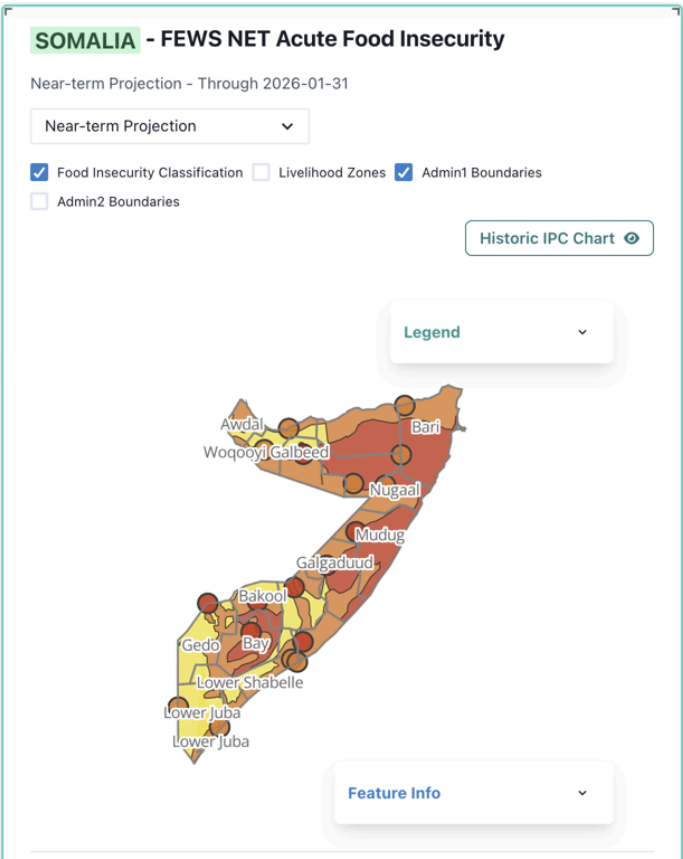
The 3 files in this folder contain API calls that are proxied through an AWS Amplify Lambda function to handle authentication and CORS. You will need to set up a similar backend proxy or modify the code to call APIs directly.

Lambda Function Endpoints:

Endpoint Path	Description
/supplyChainsApi/fews-api	Proxy for FEWS NET API calls
/supplyChainsApi/harvest-portal-api-key	Returns Harvest Portal API key
/supplyChainsApi/map-tiler-key	Returns MapTiler API key

File: AcuteFoodInsecurityMap.js

Interactive map displaying FEWS NET Acute Food Insecurity classification data with livelihood zones, administrative boundaries (Admin1/Admin2), and historical IPC classification timeline charts.



Required Libraries

```
1 {
2   "dependencies": {
3     "@chakra-ui/react": "^2.x",
4     "aws-amplify": "^6.x",
5     "maplibre-gl": "^3.x",
6     "react": "^18.x",
7     "react-icons": "^4.x",
8     "react-redux": "^8.x",
9     "recharts": "^2.x"
10  }
11 }
```

Data Sources

Resource	Source	Resource ID / Endpoint	Alternative
Admin1 Boundaries	Harvest Portal	559cbd46-5919-4ded-bd0e-e12f071e193e	https://data.harvestportal.org/dataset/fewsnet_geographicboundaries/resource/559cbd46-5919-4ded-bd0e-e12f071e193e
Admin2 Boundaries	Harvest Portal	b451bf11-ff32-4139-	https://data.harvestportal.org/dataset/fewsnet_geographicboundaries/resource/b451bf11-ff32-4139-

		9676-ef43f72b49a7	aset/fewsnet_geographicboundaries/resource/b451bf11-ff32-4139-9676-ef43f72b49a7
Livelihood Zones	FEWS NET API (via Lambda)	feature.geojson endpoint	https://help.fews.net/fde/v3/fews-net-api
IPC Phase Timeline	FEWS NET API (via Lambda)	ipcphase.json endpoint	https://help.fews.net/fde/v3/fews-net-api
Base Map Tiles	MapTiler	Requires API key	https://docs.maptiler.com/cloud/api/authentication-key/
Alternative resource for: IPC Phase Timeline only for Somalia	Harvest Portal	In place of the FEWS NET API to obtain the IPC Phase for Somalia you can use the harvest portal API to obtain the geoJSON only for the most recent classification.	https://data.harvestportal.org/dataset/acute-food-insecurity-classification/resource/f71147c0-1256-4415-9ea3-1e7fc9c20a87

API Reference

Harvest Portal API

Base URL: `https://data.harvestportal.org/api/action/`

Authentication: Requires API key passed via Authorization header

Endpoint	Description	Parameters
<code>resource_show?id={resource_id}</code>	Get resource metadata including download URL	id (Resource ID)

Admin Boundary Resource IDs:

```
1 const ADMIN1_BOUNDARY_RESOURCE_ID = '559cbd46-5919-4ded-bd0e-e12f071e193e';
```

```
2 const ADMIN2_BOUNDARY_RESOURCE_ID = 'b451bf11-ff32-4139-9676-ef43f72b49a7';
```

- ADMIN1_BOUNDARY_RESOURCE_ID:
https://data.harvestportal.org/dataset/fewsnat_geographicboundaries/resource/559cbd46-5919-4ded-bd0e-e12f071e193e
- ADMIN2_BOUNDARY_RESOURCE_ID:
https://data.harvestportal.org/dataset/fewsnat_geographicboundaries/resource/b451bf11-ff32-4139-9676-ef43f72b49a7

FEWS NET API (via Lambda Proxy)

Lambda Path: `/supplyChainsApi/fews-api`

Query Parameter	Description	Example Values
api_name	FEWS NET endpoint name	feature.geojson, ipcphase.json, ipcphase.geojson
country_code	ISO2 country code	SO, ET, KE
unit_type	Geographic unit type	livelihood_zone
as_of_date	Date filter	2015-01-01
scenario	IPC scenario	cs (Current Situation)
latlon	Latitude,Longitude	5.0,46.0
show_ipc_only	Filter to IPC data only	True
fields	Fields to return	geographic_unit_name, period_date,value,scenario,description,unit_type

Example Livelihood Zone Request:

```
1 const restOperation = get({
2   apiName: 'SupplyChainsApi',
3   path: '/supplyChainsApi/fews-api',
4   options: {
5     queryParams: {
6       api_name: 'feature.geojson',
7       country_code: 'SO', // ISO2 code
8       unit_type: 'livelihood_zone',
9       as_of_date: '2015-01-01',
10    },
11  },
12 });
```

Example IPC Phase Timeline Request:

```
1 const restOperation = get({
2   apiName: 'SupplyChainsApi',
3   path: '/supplyChainsApi/fews-api?api_name=ipcphase.json&scenario=cs&show_ipc_only=True&latlon=${lat},${lon}&fields=geographic_unit_name,period_date,value,scenario,description,unit_type',
4 });
```

Local Data Files Required

- `faostat-country-codes.json` - Country code mappings (ISO2/ISO3)

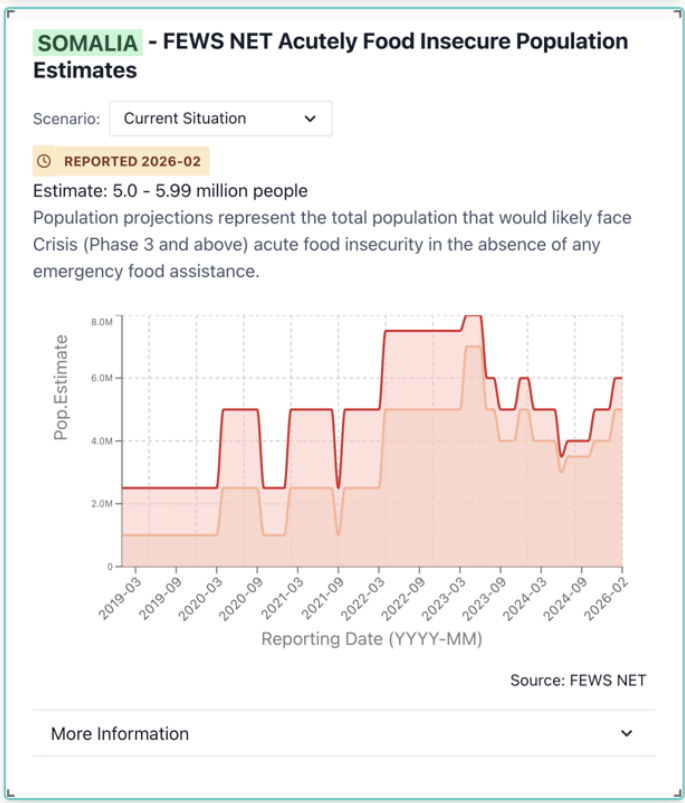
Country Resource Map

The component uses a `countryResourceMap` object that maps ISO3 country codes to Harvest Portal Resource IDs for IPC Phase GeoJSON data. Example:

```
1 const countryResourceMap = {
2   'SOM': 'f71147c0-1256-4415-9ea3-1e7fc9c20a87',
3   'ETH': 'resource-id-for-ethiopia',
4   'KEN': 'resource-id-for-kenya',
5   // ... additional countries
6 }
7
```

File: `IPCPopSizeChart.js`

Area chart displaying historical estimates of acutely food insecure population over time, with scenario selection (Current Situation vs Medium-term Projection).



Required Libraries

```
1 {
2   "dependencies": {
3     "@chakra-ui/react": "^2.x",
4     "aws-amplify": "^6.x",
5     "react": "^18.x",
6     "react-icons": "^4.x",
7     "react-redux": "^8.x",
8     "recharts": "^2.x"
9   }
10 }
```

Data Sources

Resource	Source	Endpoint
----------	--------	----------

IPC Population Size	FEWS NET API (via Lambda)	ipcpopulationsize.json
---------------------	---------------------------	------------------------

API Reference

FEWS NET API (via Lambda Proxy)

Lambda Path: /supplyChainsApi/fews-api

Query Parameter	Description	Example Values
api_name	FEWS NET endpoint name	ipcpopulationsize.json
country_code	ISO2 country code	S0 , ET , KE
scenario	IPC scenario	CS (Current Situation), ML (Medium-term Projection)
datasourcedocument	Data source document ID	6986

Example Request:

```
1  const restOperation = get({
2    apiName: 'SupplyChainsApi',
3    path: '/supplyChainsApi/fews-api',
4    options: {
5      queryParams: {
6        api_name: 'ipcpopulationsize.json',
7        country_code: 'S0', // ISO2 code
8        scenario: 'CS',
9        datasourcedocument: '6986',
10     },
11   },
12 });
```

Response Structure:

```
1  {
2    success: true,
3    value: [
4      {
5        reporting_date: "2024-01-01",
6        low_value: 1500000,
7        high_value: 2000000,
8        population_range: "1.5M - 2.0M",
9        phase_name: "Crisis (IPC Phase 3)",
10       scenario_name: "Current Situation",
11       fewsnet_region: "East Africa",
12       country: "Somalia",
13       phase: 3
14     },
15     // ... additional entries
16   ]
17 }
```

Scenario Options:

Code	Description
CS	Current Situation

ML

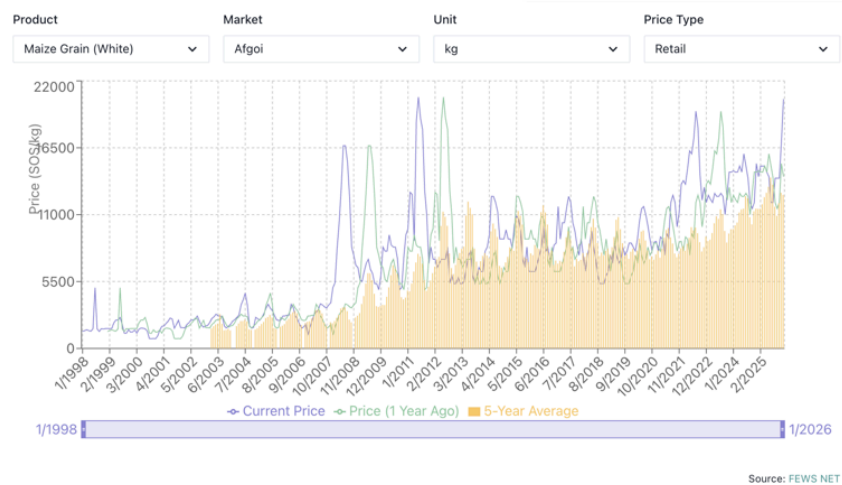
Medium-term Projection

Local Data Files Required

- `faostat-country-codes.json` - Country code mappings (ISO2/ISO3)

File: `MarketPriceFacts.js`

Composed chart (line + bar) displaying commodity price data over time with cascading filter dropdowns for product, market, unit, and price type selection. Shows current price, price from one year ago, and 5-year average.



Data Sources

Resource	Source	API Key Required
Market Price Data	Harvest Portal CKAN API	Yes

API Reference

Base URL: `https://data.harvestportal.org/api/action/`

Authentication: Requires API key obtained through `/supplyChainsApi/harvest-portal-api-key` endpoint and passed via Authorization header

Endpoint	Description	Parameters
<code>datastore_search</code> <code>sql</code>	SQL query for price data	<code>sql</code> (SQL query string)

Example SQL Queries:

Fetch available products:

```
1 SELECT DISTINCT "product"  
2 FROM "b94ba2b0-2f17-4b56-a3cd-c729dbbc8d45"  
3 WHERE "collection_status" = 'Published'  
4 ORDER BY "product"
```

Fetch price data:

```
1 SELECT *
2 FROM "b94ba2b0-2f17-4b56-a3cd-c729dbbc8d45"
3 WHERE "product" = 'Maize Grain (White)'
4     AND "unit" = 'kg'
5     AND "price_type" = 'Retail'
6     AND "collection_status" = 'Published'
7     AND "market" = 'Afgoi'
8 ORDER BY "period_date" ASC
```

Resource IDs: Each country has a unique Resource ID for market price data. The code includes edits so it only pulls data for Somalia. The Somalia resource_id is "b94ba2b0-2f17-4b56-a3cd-c729dbbc8d45"

Local Data Files Required

- `faostat-country-codes.json` - Country code mappings (ISO2/ISO3)
- Structure:

```
1 [
2   {
3     "Country": "Somalia",
4     "ISO2 Code": "SO",
5     "ISO3 Code": "SOM"
6   },
7   {
8     "Country": "Ethiopia",
9     "ISO2 Code": "ET",
10    "ISO3 Code": "ETH"
11  }
12  // ... additional countries
13
14 ]
```
